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SRM-Net: A Dual-Stream Deep Learning Framework for Localizing Digital Inpainting in Academic Credentials

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***Abstract*—Academic documents such as certificates and transcripts are majorly targeted by advanced digital forgery, raising serious concerns for institutional trust and automated verification systems. Among these, digital inpainting attacks—where original text from the documents is removed and convincingly replaced with error prone content—are especially difficult to detect because they dont leave any visible traces, even in high-resolution scans. Traditional forensic methods like Error Level Analysis (ELA), which rely on compression irregularities, often fail when applied to clean, high-quality scanned documents.**

To address these challenges, this paper presents SRM-Net, a dual-stream deep learning framework designed to detect inpainting-based forgeries and errors at the pixel level. The model combines a noise-focused stream using fixed Spatial Rich Model (SRM) filters to record subtle high-frequency residual changes, and a semantic RGB stream built on a pre-trained ResNet-18 network to learn the document's structure and visual context. Features from both streams are combined and decoded using an encoder-decoder network to generate a binary mask that localizes tampered or forged regions.

Due to the lack of publicly available academic forgery datasets, the model is trained and evaluated on a synthetically generated dataset that simulates realistic text-erasure and replacement scenarios. Experimental evaluation shows that the proposed method achieves an F1-score of 0.91, recall of 97.5%, and pixel-level accuracy of 99.8%, demonstrating that combining noise residual analysis with semantic feature learning is highly effective for detecting even visually seamless inpainting attacks in scanned academic documents.

***Index Terms*—Image Forensics, Document Forgery Detection, Spatial Rich Model (SRM), Dual-Stream Neural Network, Digital Inpainting, Residual Noise Analysis, Academic Credential Verification, Deep Learning.**

I. Introduction

Documents like certificates, grade sheets, and mark sheets are considered to be primary proof of academic credentials and can easily be verified/used during the time of admissions, recruitment, or evaluation. Over the years, most of these papers have gone through the process of digitalization. Although digitalization is a great advantage in itself, there also exist threats of manipulations in these documents due to advancements in image manipulation software.

Digital inpainting is one of the toughest techniques for document manipulation. In this attack, original text information, like the names of students or their grades, is carefully erased and replaced with new information. The inpainting technique is used to smoothen the area around the erased information so that there are no gaps, and the document is very hard to detect. Thus, the forged documents become so real that they can dupe experts.

Conventional forensic analysis of documents is mostly centered on detecting pixel errors caused by JPEG compression. Other tools like Error Level Analysis look at the difference in JPEG compression levels to detect manipulated portions of an image. While these tools work well when the image is of poor quality due to high JPEG compression Ratios, they fail when used on high-quality scanned documents. Here, the compression levels are insignificantly small; hence the forged portion will smoothly blend with the rest of the image.

With the accelerated pace of advancements in deep learning, convolutional neural networks have been used for successful applications in the field of image analysis, including object detection and segmentation. The models have the ability to train for spatial patterns using the data. However, traditional models which use the information available in the RGB images are biasing more towards the semantic information, which sometimes might not be adequate for the detection of finer details in the process of inpainting. In fact, the process of inpainting has more effects on the noise patterns at the higher frequency levels.

Inspired by this, this paper proposes **SRM-Net**, which is a dual-stream deep learning architecture for digital inpaint-

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ing attack detection in academic documents. The proposed architecture incorporates two different sources of information. The first incorporates noise residue analysis via Spatial Rich Model (SRM) filters to identify the introduced smooth artifacts in the residual noise. The second incorporates RGB image features via a convolutional neural network architecture. By combining features of both sources, the architecture learns a better representation of forged areas.

For verification purposes, a set of experimental results is carried out using a synthetic database created to stimulate realistic scenarios of text removal and text replacement in document forgery. The experimental results show the effectiveness of the proposed method in localizing the areas of the text-inpainted regions in high-quality scans of academic documents.

II. RELATED WORKS

Computer vision-based image forensics has also received considerable attention in the last few years, especially in the context of image tampering and document alteration. Some of the initial studies performed in this domain were related to the detection of visible tampering in image processing techniques like copy-move, image splicing, and image resampling.

Among them, the approaches based on compression forensics have represented the most widely used methods for the detection of forgery. ELA is a representative technique that localizes tampered regions by considering the differences in JPEG compression artifacts. This kind of techniques might perform well for either low-quality images or web-compressed content, but their reliability drops significantly when dealing with high-quality scanned documents where compression artifacts may either be uniform or negligible. For this reason, subtler manipulations like digital inpainting remain problematic for compression-dependent methods.

Noise-based analysis has also found applications in image forensics. It has been shown in past studies that the noise present in authentic images is usually disrupted by the manipulation tasks performed on the image. The idea of Spatial Rich Model (SRM) features was first considered for image steganalysis to quantify the noise residues in the image. However, it has also found applications in forgery detection. It is effective in analyzing the noise removal effect of surface smoothing, which is caused by image inpainting. [1].

With the rise of deep learning, the use of convolutional neural networks in forgery localization tasks has gradually gained momentum. Deep learning models, specifically CNN, have proved effective in discriminative feature learning from raw images, allowing for the localization of forgeries at the pixel level. Encoder-decoder models have particularly gained popularity for their potential to maintain spatial resolution when doing dense regression. Nonetheless, the majority of the existing models utilizing deep learning techniques solely focus on RGB cues, leaving their susceptibility to noise-level forgeries unaddressed.

Recent studies have explored hybrid approaches that combine multiple feature representations to improve detection robustness. Motivated by this direction, the present work integrates noise residual analysis with semantic feature learning by employing a dual-stream architecture. By combining SRM-based noise features with RGB-based contextual features extracted using a convolutional backbone, the proposed approach aims to overcome the limitations of both traditional forensic methods and single-stream deep learning models.

Some recent deep learning techniques have achieved better performance in forgery localization through learning a more complex hierarchy using feature learning techniques. For example, Zhou et al. proposed feature learning-based manipulation detection framework methods that improve the representation of spatial features [2]. More recent studies have investigated the utilization of CNNs combined with transformer networks for better robustness in localization [3].

III. PROPOSED METHODOLOGY

This section presents the proposed SRM-Net framework developed for detecting digital inpainting attacks in academic documents. Overall, the model should achieve an accurate localization of forged regions at the pixel level by jointly analyzing semantic visual information and noise-level inconsistencies introduced during manipulation.

A. Overall Architecture

The architecture of the proposed system is dual-stream encoder–decoder based; see Fig. 1. Here an RGB document image of size 256×256 is fed into the model for processing through two parallel streams of feature extraction. The first stream focuses on analyzing noise residual using SRM filters, while the second stream aims at extracting semantic and structural features from the RGB image through a CNN backbone. Features from both streams are fused and passed through a decoder network that generates a binary segmentation mask indicating forged and non-forged regions.

B. Noise Stream Based on SRM Filters

The noise stream is designed to capture subtle artifacts introduced by digital inpainting operations. Inpainting typically involves surface smoothing and noise suppression to remove visible discontinuities, which alters the natural noise distribution of the document image. To capture these artifacts, the input RGB image is first passed through a set of non-trainable Spatial Rich Model (SRM) filters.

SRM filters act as high-pass filters that emphasize noise residuals while suppressing semantic content. Unlike trainable convolutional filters, SRM filters are fixed and are specifically designed to highlight inconsistencies in local pixel correlations. The output of the SRM layer is then processed through a sequence of convolutional blocks consisting of convolution, batch normalization, and activation functions to progressively downsample the feature maps and extract higher-level noise representations.

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TABLE I
COMPARISON OF EXISTING DOCUMENT FORGERY DETECTION APPROACHES

Method	Feature Type	Detection Technique	Strengths	Limitations
Error Level Analysis (ELA)	Compression Artifacts	Statistical analysis of JPEG re-compression	Simple to implement; effective on low-quality images	Fails on high-quality scans; dependent on compression artifacts
Handcrafted Forensic Methods	Noise, Edges, Texture	Statistical feature extraction	Interpretable features; low computational cost	Limited robustness; poor generalization to complex forgeries
CNN-based RGB Models	RGB Visual Features	Deep learning classification or segmentation	Strong semantic feature learning; end-to-end training	Insensitive to subtle noise-level manipulations such as inpainting
Noise-based Methods (SRM)	Noise Residuals	High-frequency residual analysis	Effective in detecting smoothing and noise suppression artifacts	Lack semantic context; limited standalone performance
Proposed SRM-Net	Noise + RGB Features	Dual-stream CNN with feature fusion	Robust to high-quality scans; accurate pixel-level localization	Requires training data; higher computational cost

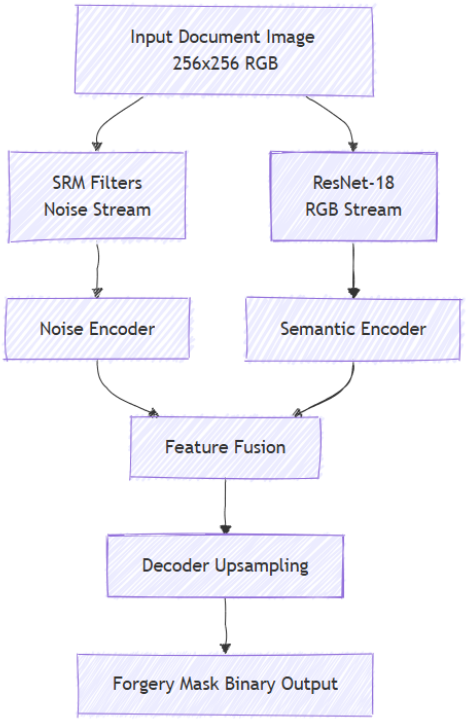


Fig. 1. Overview of the proposed SRM-Net architecture. The model consists of a noise stream based on SRM filters and an RGB stream for semantic feature extraction. Fused features are decoded to produce a pixel-level forgery localization mask.

C. RGB Stream for Semantic Feature Extraction

The RGB stream involves extracting semantic and structural information from a document image. The RGB stream uses a ResNet-18 model as its backbone. This ResNet-18 model is pre-trained on the ImageNet dataset. The usage of a pre-trained network helps the model in understanding features associated with text structure, characters, and a document layout.

The input RGB map is processed through the convolutional layers in the ResNet-18 architecture, and the intermediate feature maps are obtained from the encoder. The features obtained include contextual features, which are necessary for distinguishing between the real document region and the region with forged or replaced text information.

D. Feature Fusion Strategy

After the feature extraction step, the outputs from the noise stream and the RGB stream undergo a process where they are aligned and concatenated along the channel dimension. This approach allows the network to scrutinize the residual patterns for noise as well as the visual semantic information simultaneously, hence making the network sensitive to the inpainting-based forgeries that the single feature representation may not capture.

The combined feature maps are then used as input to the decoder neural network that does pixel-level classification.

E. Decoder and Forgery Localization

The neural network decoder is developed on an architecture that employs upsampling during the reconstruction of the spatial resolution in the input image. It consists of a sequence of layers comprising upsampling followed by a convolutional operation that smooths out the fusion features for the recovery of fine-grained details. The last series applies the sigmoid function when creating a binary probability map using pixel values closest to one in forged areas and zero in authentic areas.

It is to be noted that the resolution of the output segmentation mask is the same as that of the image, thus giving an accurate localization of the inpainted area. Such property of the output makes this proposed approach useful for real-world document verification, for which the identification of the exact position of forgery is important.

IV. DATASET AND EXPERIMENTAL SETUP

A. Dataset Preparation

A synthetic dataset was created for this study because there are no publicly accessible datasets that specifically address inpainting-based forgery detection in academic documents. About 3,000 document images that are intended to mimic actual scanned academic certificates in terms of background texture, font style, and noise characteristics make up the dataset.

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Noisy white backgrounds with printed text, including names, grades, and identification numbers, were used to create authentic samples. Forged samples were produced by mimicking digital inpainting attacks, in which new text was added after specific text regions were eliminated using smoothing techniques. This method closely resembles actual document tampering scenarios by introducing subtle noise inconsistencies while maintaining visual continuity.

For each forged document image, a corresponding binary mask for the ground truth has been created to point out the location where manipulation is done. The forged pixels should be marked as one, and others should be zero. The size of all images is been set to a fixed value of 256×256 pixels before the training process.

B. Training, Validation, and Test Split

A split ratio of 70-15-15 was adopted to divide the data into the training, validation, and testing sets respectively. The validation set was utilized to monitor performance and for hyperparameter tuning, and the testing set was reserved to determine generalization abilities, and so was not modified.

C. Training Setup

Using paired document images and matching ground-truth masks, the suggested SRM-Net model was trained under supervision. Pixel-level classification was optimized using binary cross-entropy loss as the objective function. Standard data augmentation methods, such as random cropping and slight intensity variations, were used to increase robustness and decrease overfitting after the Adam optimizer was used for training.

D. Evaluation Metrics

To measure the performances of these models, the standard evaluation metrics used for segmentation: pixel accuracy, precision, recall, and F1 values, are used. Pixel accuracy measures how many pixels are correctly classified, while the values of precision and recall measure the efficacy of the model to locate the forged region appropriately.F1 measures the efficacy of either precision or recall, which is more applicable when there is class imbalance concerning the location of the forgery region.

E. Implementation Details

All the experiments were done in Python using the PyTorch deep learning framework. Standard image processing libraries were used for image preprocessing and mask generation. The model was trained and evaluated on a local development environment with GPU acceleration; the performance on the validation set was monitored to ensure stable convergence during training. sectionResults and Discussion

F. Performance Evaluation

The ability of the suggested SRM-Net framework to identify and locate inpainting-based forgeries in academic documents was assessed using the synthetic test dataset. Standard segmentation metrics that assess both localization quality and detection accuracy were used in the evaluation.

TABLE II
PERFORMANCE OF SRM-NET ON THE SYNTHETIC TEST DATASET

Metric	Value
Pixel Accuracy	99.8%
Precision	0.86
Recall	97.5%
F1-Score	0.91



Fig. 2. Visual comparison of forgery detection results. From left to right: input forged document, baseline Error Level Analysis (ELA) output, and the proposed SRM-Net result. The proposed method successfully localizes the inpainted grade region, whereas the ELA method fails to provide clear localization.

Table II presents the quantitative performance of the suggested model: With a pixel accuracy of 99.8%, most document pixels are correctly classified by the SRM-Net. A recall of 97.5% demonstrates the strong ability of the model to detect forged regions, which is crucial for document verification scenarios. An F1-score of 0.91 has been obtained to confirm that robustness of the suggested method, showing a balanced trade-off between precision and recall.

Although this was not performed from a quantitative perspective due to computation constraints, qualitatively from the development process of the model, the former configuration appeared to focus mainly on structural/textual boundaries, with the latter configuration being more sensitive to noise inconsistencies despite lacking any form of understanding of the contextual data. The combination of the two streams always appeared to provide more stability with the predictions.

G. Visual Analysis

A qualitative comparison of a forged academic document, the output from a conventional Error Level Analysis (ELA) method, and the outcome from the suggested SRM-Net model is shown in Figure 2. A manipulated grade region made with digital inpainting techniques can be found in the forged document.

The ELA-based method is unable to precisely identify the forged area in the high-quality scanned document, as shown in the figure. Poor localization is caused by the weak and dispersed compression artifacts. The SRM-Net output, on the other hand, allows for accurate forgery localization by displaying strong activation in the manipulated region while suppressing responses in genuine areas.

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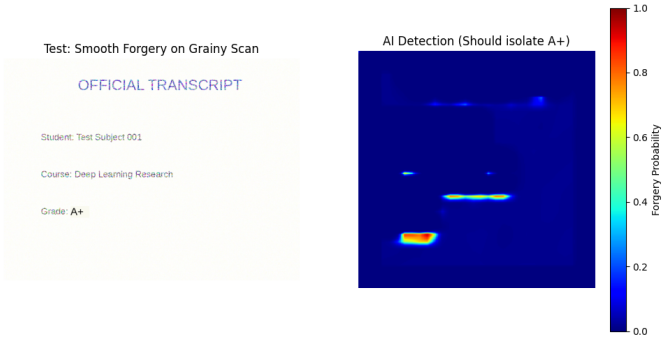


Fig. 3. Robustness check on a noisy scanned document: the left image represents a forged academic document with additional noise, while the right image is the forgery probability heatmap generated by SRM-Net. It can be seen that the model still successfully segmented the manipulated grade region even though the background noise was enhanced.

H. Observations

When applied to high-quality scanned academic documents, the experimental results demonstrate the limitations of conventional compression-based forensic techniques. The suggested SRM-Net framework successfully captures minute artifacts brought about by digital inpainting operations by combining semantic feature extraction from RGB images with noise residual analysis using SRM filters. The results show great potential for real-world academic document verification, despite the fact that the current evaluation is based on a synthetic dataset. In the future, the dataset will be expanded to include actual scanned documents, and generalization across various document layouts and scanning conditions will be improved.

V. Conclusion and Future Work

SRM-Net, a dual-stream deep learning framework for identifying and locating digital inpainting forgeries in scholarly documents, was introduced in this paper. In order to jointly capture high-level structural features introduced during manipulation and subtle noise residual inconsistencies, the suggested method combines an RGB-based semantic stream with an SRM-based noise stream. The framework overcomes the drawbacks of conventional compression-dependent forensic methods and permits precise pixel-level forgery localization by combining these complementary representations. Strong performance was shown in the experimental evaluation on a synthetically generated dataset, with high recall and overall pixel accuracy. The suggested method effectively isolates manipulated regions in high-quality scanned documents where traditional techniques like Error Level Analysis are unable to provide accurate localization, according to qualitative analysis.

Although it must be noted that even this assessment is executed on a precision-crafted synthetic dataset that emulates actual scenarios of inpainting, practical academic documents may contain different levels of variation, such as printed artifacts, illumination conditions, etc., and layout structures. Hence, future work will focus on increasing the dataset to improve generalization on various documents, as well as validating the proposed framework on actual scanned documents. Efficient deployment of it in actual practice for academic document verification systems may also be achieved by further work to explore different variants of this model.

Acknowledgment

The preferred spelling of the word “acknowledgment” in America is without an “e” after the “g”. Avoid the stilted expression “one of us (R. B. G.) thanks . . .”. Instead, try “R. B. G. thanks. . .”. Put sponsor acknowledgments in the unnumbered footnote on the first page.

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